Loop – repetition of task



What is Loop?

- There are many situation where we need to repeat same set of tasks again and again. Like while writing table of any number we multiply same number from 1 to 10, multiplying number from n to 1 to find the factorial.

Python Loop Statements

- To carry out repetition of statements Python provide 2 loop statements
  - Conditional loop (while)
  - Counting loop (for)

range() function

- Before we proceed to for loop let us understand **range()** function which we will use in for loop to repeat the statement to **n number** of times.
- Syntax:
  - range(lower_limit, upper_limit)
- The range function generate set of values from **lower_limit** to **upper_limit- 1**

range() function

- For e.g.
- **range(1,10)** will generate set of values from **1-9**
- range(0,7) will generate [0-6]
- Default step value will be +1 i.e.
- range(1,10) means (1,2,3,4,5,6,7,8,9)

range() function

- To change the step value we can use third parameter in range() which is step value
- For e.g.
- **range(1,10,2)** now this will generate **value [1,3,5,7,9]**
- Step value can be in –ve also to generate set of numbers in reverse order.
- range(10,0,-1) will generate number as [10,9,8,7,6,5,4,3,2,1]

range() function

- To create list from 0 we can use
- **range(10)** it will generate **[0,1,2,3,4,5,6,7,8,9]**

for loop

- for loop in python is used to create a loop to process items of any sequence like List, Tuple, Dictionary, String
- It can also be used to create loop of fixed number of steps like 5 times, 10 times, n times etc using **range()** function.

Example – for loop with List

```
School=["Principal","PGT","TGT","PRT"]
for sm in School:
        print(sm)
```

Example – for loop with Tuple

```
Code=(10,20,30,40,50,60)
for cd in Code:
        print(cd)
```

for loop with string

```
for ch in“Plan”
        print(ch)
```

The above loop product output

P

l

a

n

Program to print table of any number

```
num = int(input("Enter any number "))
for i in range(1,11):
        print(num,'x',i,'=',num*i)
```

Program to find the sum of all number divisible by 7 between 1 to 100

```
sum=0
for i in range(1,101):
        if i % 7 == 0:
                sum+=i
print("total of number divisible by 7 between 1 to 100 is ",sum)
```

PROGRAMS

1. WAP to enter any number and find its factorial
2. WAP to print the following fibonacci series

   0, 1, 1, 2, 3, 5, 8, ……..n terms
3. WAP to enter 10 number and find the sum and average.
4. WAP to enter Lower_Limit, Upper_Limit and find the sum of all odds and evens number between the range separately

while loop

- While loop in python is conditional loop which repeat the instruction as long as condition remains true.
- It is entry-controlled loop i.e. it first check the condition and if it is true then allows to enter in loop.
- while loop contains various loop elements: **initialization**, **test condition**, **body of loop** and **update statement**

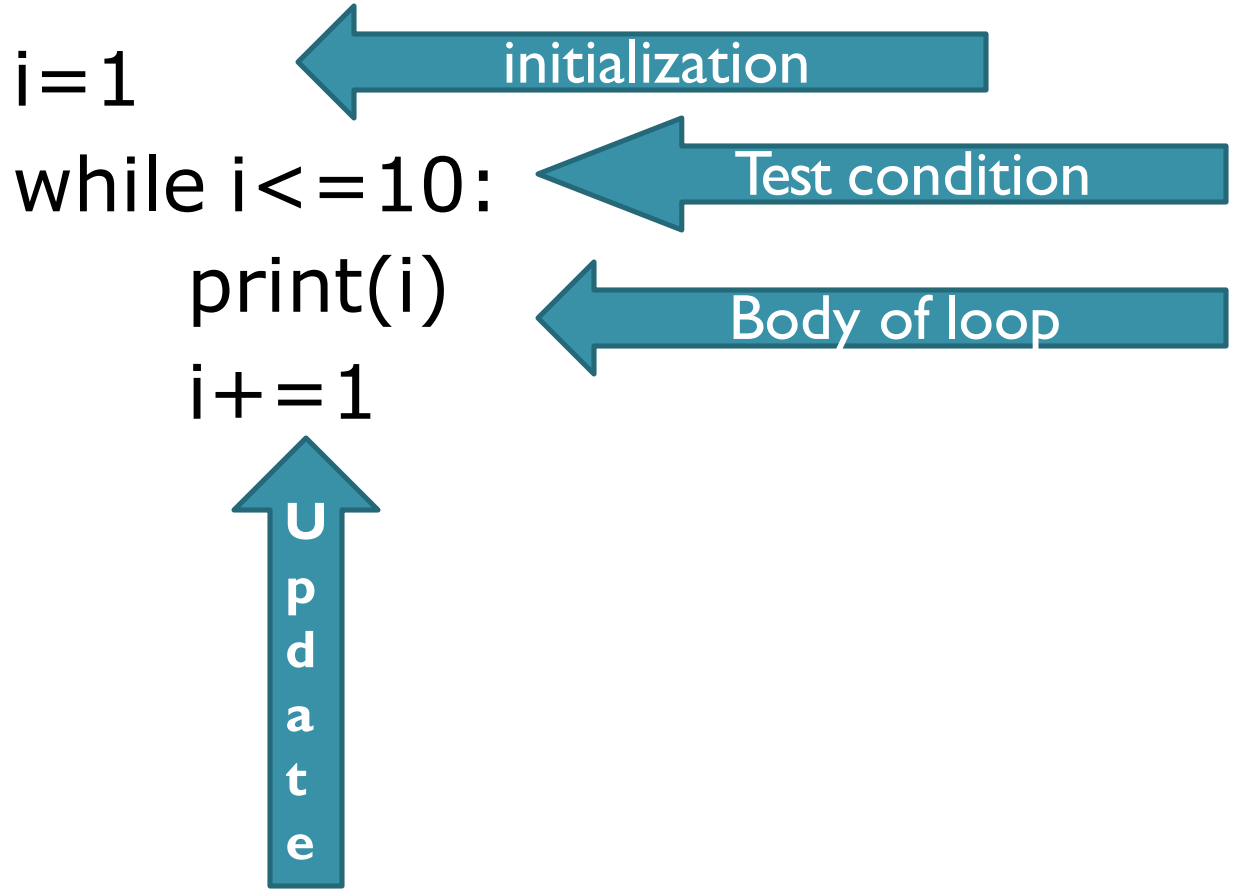
while loop elements

1. **Initialization :** it is used to give starting value in a loop variable from where to start the loop
2. **Test condition :** it is the condition or last value up to which loop will be executed.
3. **Body of loop :** it specifies the action/statement to repeat in the loop
4. **Update statement :** it is the increase or decrease in loop variable to reach the test condition.

Example of simple while loop

```
i=1
while i<=10:
        print(i)
        i+=1
```

Example of simple while loop



Programs of while loop

- WAP to enter any number and find its reverse
- WAP to enter any number and a digit and count how many times digit is in the number
- WAP to enter any number and check it is armstrong or not
- WAP to enter any number and check it is palindrome or not

WAP to enter any number and find its reverse

```
num = int(input("Enter any number "))
rev = 0
num2=num
while num>0:
        rem = num % 10
        rev = rev * 10 + rem
        num //=10
print("Reverse of ",num2," is ",rev)
```

Jump Statements – **break** & **continue**

break statement in python is used to **terminate the containing loop** for any given condition. Program resumes from the statement immediately after the loop

Continue statement in python is used to **skip the statements below continue statement** inside loop and **forces the loop to continue with next value**.

Example – break

```
for i in range(1,20):
        if i % 6 == 0:
                break
        print(i)
print("Loop Over")
```

The above code produces output

1
2
3
4
5
Loop Over

when the value of i reaches to 6 condition will becomes True and loop will end msg Loop Over will be printed

Example – continue

```
for i in range(1,20):
        if i % 6 == 0:
                continue
        print(i, end=“)
print(“Loop Over”)
```

The above code produces output

1 2 3 4 5 7 8 9 10 11 13 14 15 16 17 19

Loop Over

when the value of i becomes divisible from 6, condition will becomes True and loop will skip all the statement below continue and continues in loop with next value of iteration.

Loop .. else ..Statement

- Loop in python provides else clause with loop also which will execute when the loop terminates normally i.e. when the test condition fails in while loop or when last value is executed in for loop **but not when break terminates the loop**

Syntax of Python loop along with else clause

for with "else"	while with "else"
for <var> in <seq>: statement 1 statement 2 **else:** statement(s)	while <test condition>: statement 1 statement 2 **else:** statement(s)

Example ("else" with while)

```
i=1
while i<=10:
        print(i)
        i+=1
else:
        print("Loop Over")
```

Output

```
1
2
3
4
5
6
7
8
9
10
Loop Over
```

Example (“else” with for)

```
names=["allahabad","lucknow","varanasi","kanpur","agra","ghaziabad"
,"mathura","meerut"]
city = input("Enter city to search: ")
for c in names:
        if c == city:
                print(“City Found")
                break
else:
        print("Not found")
```

Output

Enter city to search :varanasi
City Found
Enter city to search :unnao
Not found

Example- program to input number and tests if it is a prime number or not

```
num = int(input("Enter any number :"))
lim = num//2 + 1
for i in range(2,lim):
        rem = num % i
        if rem == 0:
                print(num," is not prime")
                break
else:
        print(num," is a prime number ")
```

Example- program to input number and tests if it is a prime number or not - **OUTPUT**

```
Enter any number :23
23  is a prime number
```

```
Enter any number :36
23  is not a prime number
```